BEING IND-QUASI-AFFINE IS STABLE UNDER BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 19639-19652, immediately following the statement at 19634-19637.

Editorial note. The author's proof is a reduction through four cited chapter lemmas; all four are navigation pointers into the same chapter and carry no part of the statement being counted.

Editorial note. The statement itself is short but is not restated anywhere else in this chapter or in its published ranges.

Allowed standard prerequisites (editorial).

- Alternative definition of ind-quasi-affine. chapter lemma lemma-ind-quasi-affine-alternative-definition, cited at proof line 19642. Lets the property be checked Zariski locally on the target, so the base may be assumed affine.
- Composition of ind-quasi-affine morphisms. chapter lemma lemma-ind-quasi-affine-composition, cited at proof line 19647. Combines the affine base change X x_Y Z -> X with the given morphism X -> Y.
- Ind-quasi-affine examples, including affine morphisms. chapter lemma lemma-ind-quasi-affine-examples, cited at proof line 19648. Supplies that the affine morphism X x_Y Z -> X is ind-quasi-affine.
- Permanence of ind-quasi-affineness. chapter lemma lemma-ind-quasi-affine-permanence, cited at proof line 19650. Transfers the property from the composite X x_Y Z -> Y to the base change X x_Y Z -> Z.

Source reference. Original lemma 0F1U: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19538–19538. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0F1V: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19585–19585. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0F1W: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19606–19606. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0F1X: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19621–19621. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *The property of being ind-quasi-affine is stable under base change.*

Proof. Let $f : X \to Y$ be an ind-quasi-affine morphism. For checking that every base change of f is ind-quasi-affine, by Lemma 0F1U, we may work Zariski locally on Y, so we assume that Y is affine. Furthermore, we may also assume that in the base change morphism $Z \to Y$ the scheme Z is affine, too. The base change $X \times_Y Z \to X$ is an affine morphism, so, by Lemmas 0F1V and 0F1W, the map $X \times_Y Z \to Y$ is ind-quasi-affine. Then, by Lemma 0F1X, the base change $X \times_Y Z \to Z$ is ind-quasi-affine, as desired. □